

Assignment 1 MATLAB

1. (a.) -2^3+9 or $-(2)^3+9$ or $-(2^3)+9 = 1$
(b.) $2/3*3=2$
(c.) $3*(2/3)=2$
(d.) $3 * 4 - 5^2 * 2 - 3=-41$
(e.) $(2/3^2)*(3-4^3)^2= 826.8889$
(f.) $3*(3*4-2*5^2-3)=-123$
2. (a.) $x=(2+6i)/(9i)*(1+i)=0.8889 + 0.4444i$
(b.) $\text{abs}(x)= 0.9938$, $\text{angle}(x)= 0.4636$ or $\text{atan}(\text{imag}(x)/\text{real}(x))$
3. $A = [2, 4, 1; 6, 7, 2; 3, 5, 9]$
(a.) `>> x=A(1,:)`
(b.) `>> y=A(2:3,:)`
(c.) `>> sum(A(:,1:3))`
4. `>> X = [3, 2, 6, 8]'`
`>> Y = [4, 1, 3, 5]'`
`>> X+Y`
`>> sum(X+Y)= 32`
5. `>> A = [1, 2, 3, 4; 5, 6, 7, 8; 1, 1, 2, 3]`
(a.) `>> B=A([1 3],[2 4])`
(b.) `>> A([1 3],:)=A([3 1],:)`
(c.) `>> A(:)` a column vector
(d.) `>> A(:,2)=[ ];`
`>> A(3,:)= [ ];`
`>> A`
(e.) `>> A(:,2)= [ ]`
6. `>> a = [2,-1, 5, 0];`
`>> b = [3, 2,-1, 4];`
(a.) `>> c = a-b = -1 -3 6 -4`
(b.) `>> c = +a-3*b = -7 -7 8 -12`
(c.) `>> c = 2*a + a.^b = 12.0000 -1.0000 10.2000 0`
(d.) `>> c = b./a = 1.5000 -2.0000 -0.2000 Inf`
(e.) `>> c = b.*a = 6 -2 -5 0`
(f.) `>> c = 2*b/3.*a = 4.0000 -1.3333 -3.3333 0`
(g.) `>> c = b*2.*a = 12 -4 -10 0`
7. `>> [x, y] = meshgrid (-8 : 0.5 : 8);`
`>> r = sqrt( x.^2 + y.^2) + eps;`
`>> z = sin(r)./r;`
`>> mesh(z)`
8. `>> 2*(3+4)/(5*(6 + 1))^2 =0.0114`

9.

(a.) `>> 1+[2, 3,-1] or >> 1+[2, 3,-1]`

(b.) `>> [1, 4, 8].*3 or >> [1, 4, 8]*3`

(c.) `>> [1, 2, 3].*[0,-1, 1]`

(d.) `>> [2, 3, 1].^2`